

Technology Stack

Date	28 June 2026
Team ID	
Project Name	Rising water
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript, Jinja2	HTML, CSS, and vanilla JS were chosen for a lightweight, fast, and fully responsive "Glassmorphism" UI without relying on heavy frameworks. Jinja2 handles dynamic server-side rendering of data (like the prediction history table).
2	Backend / Server-Side	Python, Flask, XGBoost, Scikit-learn	Python provides the strongest ecosystem for Machine Learning. Flask is used as a lightweight, fast web framework. XGBoost handles the high-performance classification, and Scikit-learn provides robust data preprocessing (StandardScaler).
3	Database / Data Storage	PostgreSQL, SQLAlchemy (ORM)	PostgreSQL (via Neon.tech) provides highly reliable, scalable relational data storage for user credentials and prediction history. SQLAlchemy ORM ensures secure, injection-proof database interactions.
4	Cloud / Hosting / Deployment	Render	Render provides seamless, cost-effective, and fully managed cloud hosting with continuous deployment capabilities, ensuring the ML web application is highly available on the internet.
5	Version Control & CI/CD	Git, GitHub	Git and GitHub provide essential version control, enabling seamless code tracking, secure repository management, and direct integration with the Render deployment pipeline.

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6	Third-Party APIs / Other Tools	Werkzeug, Flask-Caching, Flask-Compress	Werkzeug provides secure pbkdf2:sha256 password hashing. Flask-Caching and Compress are utilized to optimize backend performance and minimize response payload sizes for fast page loading.